

Optimal Selection of Urban Street Trees for Ecological Value: A Case Study of Xiong'an New Area

Yufeng Liu¹, Kai Yan¹ and Aming You¹

¹ State Key Laboratory of Remote Sensing and Digital Earth, Advanced Interdisciplinary Institute of Satellite Applications, Faculty of Geographical Science, Beijing Normal University, Beijing, China, 100875.
kaiyan@bnu.edu.cn

Abstract. To improve the ecological suitability of roadside greening in newly developed urban areas, this study takes Xiong'an New Area as the case study and selects 25 common street tree species. An integrated framework was established that combines road functional gap identification, ecological constraint grading, land-use scenario coupling, and tree-species optimization. Based on multi-source data, including remote sensing, meteorological, soil, and plant trait data, differentiated road demands were identified in terms of greening, carbon sequestration, pollution mitigation, stormwater regulation, and cooling. The AHP–entropy weighting–TOPSIS method was then applied to rank candidate species comprehensively. The results show significant spatial heterogeneity in road ecological demands, with Class F and Class E roads accounting for the largest proportion. Although dominant species overlap across road categories, their functional strengths differ. Specifically, *Diospyros lotus*, *Ulmus glabra*, *Alnus incana*, and *Carya glabra* perform best in temperature reduction, carbon sequestration, pollution mitigation, and stormwater regulation, respectively. These findings suggest that street-tree optimization should follow a classified configuration strategy based on road ecological gaps, environmental constraints, and spatial scenarios.

Keywords: Xiong'an New Area; street trees; ecological optimization; road classification; tree species selection; TOPSIS.

1 Introduction

Rapid global urbanization, while driving economic growth, has also intensified environmental challenges such as the urban heat island effect, air pollution, and altered hydrological cycles. In this context, urban street trees are recognized as a core component of green infrastructure, providing critical ecosystem services (ES), including carbon sequestration, pollutant removal, stormwater regulation, and microclimate improvement. Quantifying these ecological benefits is essential for developing sustainable strategies to mitigate the adverse impacts of high-density urbanization.

Research on the carbon sequestration and ecological value of street trees has evolved from early qualitative recognition to precise quantitative assessment, and more recently, to complex system optimization. In the 1980s, Spirn's systematic

framework for urban ecology [1] and Dorney et al.'s pioneering work on urban tree biomass [2] shifted academic focus from traditional horticultural aesthetics to ecological function. By the mid-1990s, this shift prompted early quantitative efforts, such as the Chicago Urban Forest Climate Project [3], which established empirical links between urban forest structure and function [4]. The signing of the Kyoto Protocol further directed research toward the potential of trees as carbon sink assets [5]. From the late 1990s to early 2000s, in order to overcome the limitations of single-point observations, Nowak's team at the USDA Forest Service developed flagship models such as UFORE (precursor to i-Tree Eco) [6] and STRATUM (precursor to i-Tree Streets) [8], based on forestry allometric equations [9]. Case studies in locations such as Brooklyn validated the feasibility of standardized, monetized assessments of urban forest carbon stocks and ecosystem benefits [7], leading to the global adoption of these methods, including in China [10]. Since 2010, the integration of airborne LiDAR and high-resolution remote sensing has helped overcome the spatiotemporal limitations of ground surveys, significantly improving species classification and carbon mapping accuracy. The proliferation of UAVs has further enhanced micro-scale monitoring capabilities. As a result, research focus has shifted from status assessment to planning optimization. Scholars have introduced multi-objective optimization (MOO) frameworks, using evolutionary algorithms such as NSGA-II to address trade-offs among carbon sequestration, cooling benefits, and maintenance costs, while also incorporating climate resilience to adapt to future climate change.

This study takes the Xiong'an New Area as its research area, focusing on the enhancement of ecological functions and the optimal allocation of tree species in urban road spaces. Twenty-five common street tree species adapted to the local environment were selected as the subjects of this study. By integrating multi-source data—including meteorological, soil, and remote sensing environmental variables, as well as functional traits of tree species—this study first constructs a framework for identifying road ecological needs. It characterizes functional gaps in different road segments across dimensions such as greening, carbon sequestration, pollution control, stormwater management, and cooling, while using factors such as PET, soil pH, and salinity to characterize ecological constraints. Building on this foundation, the study further incorporates land-use scenario coupling, suitable tree species screening, and the AHP – Entropy Weighting – TOPSIS integrated decision-making method to quantitatively rank and select candidate tree species for different road categories. Finally, by integrating scenario simulation results for high-ranking tree species, this study evaluates their optimization potential in terms of temperature mitigation, PET reduction, NPP enhancement, and pollution reduction, thereby proposing differentiated tree species configuration recommendations tailored to different road types. This paper aims to provide quantitative evidence for the refined configuration of roadside greening in the Xiongan New Area and offers a methodological reference for the multi-objective-oriented optimization of street tree selection in new urban areas.

2 Materials and Methods

2.1 Research Area

The Xiong'an New Area ($38^{\circ}43'N$ – $39^{\circ}10'N$, $115^{\circ}38'E$ – $116^{\circ}20'E$) is located in the central North China Plain. It experiences a warm temperate, semi-humid continental monsoon climate, with annual precipitation concentrated in the flood season and high evapotranspiration rates. The region has moderate average annual temperatures and relatively long frost-free periods. Soils are mainly loess and alluvial, while lowland and lakeshore areas show potential risks of salinization. Prevailing winter winds come from the north, and summer winds mainly originate from the southeast. Dust and ozone pollution display pronounced seasonal variation.

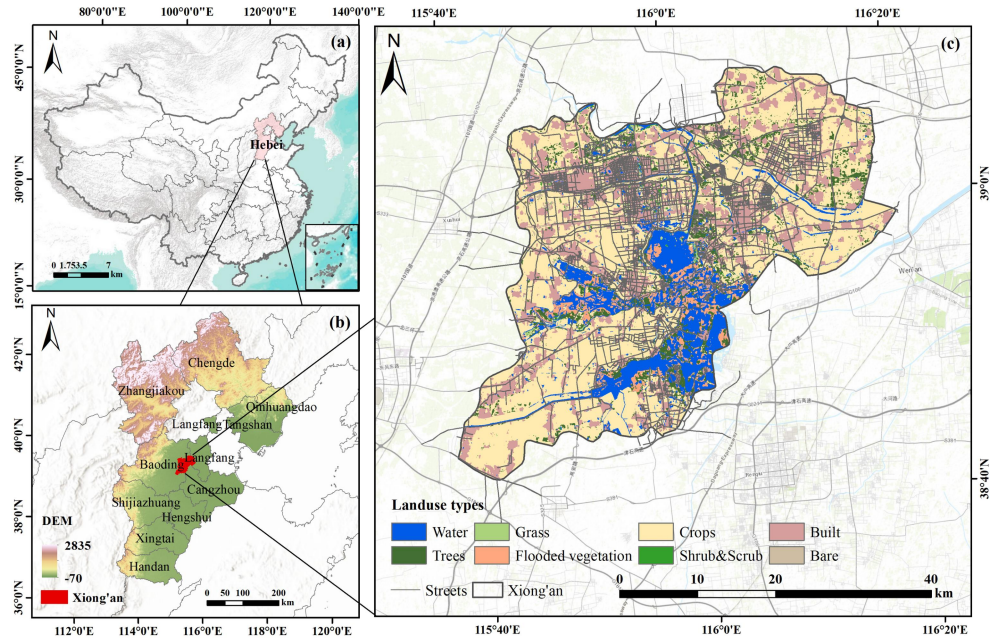


Fig 1. Research Area

2.2 Data Source

Basic Environmental Data. The data used in this study are primarily derived from publicly available remote sensing and reanalysis datasets on the Google Earth Engine platform. Specifically, land surface temperature (LST) data were obtained from the Landsat 8 LANDSAT/LC08/C02/T1_L2 dataset; the thermal infrared band ST_B10 was extracted to obtain annual 30-meter resolution LST information for the Xiong'an New Area; Net primary productivity (NPP) data were sourced from the COPERNICUS/S2_SR_HARMONIZED dataset. Annual and growing season NPP at 10-meter resolution were derived using vegetation indices, while solar radiation data were provided by the ECMWF/ERA5_LAND/DAILY_AGGR dataset; Soil organic

carbon (SOC) data are sourced from the SoilGrids database (projects/soilgrids-isric/soc_mean), with the soc_0-5cm_mean band extracted to characterize soil organic carbon content in the 0 – 5 cm surface layer; Land use data are sourced from the Google Dynamic World dataset (GOOGLE/DYNAMICWORLD/V1) and are used to generate annual 10-meter resolution land use major classification maps for the Xiong'an New Area; Among the PM_{2.5}-related variables, the raw aerosol and pollutant data are sourced from the COPERNICUS/S5P/NRTI/L3_AER_AI and COPERNICUS/S5P/NRTI/L3_NO2 datasets, respectively. Auxiliary variables include land cover data from ESA/WorldCover/v100, NDVI generated from COPERNICUS/S2_SR_HARMONIZED, NOAA/VIIIRS/DNB/MONTHLY_V1/VCMSLCFG nighttime light data, USGS/SRTMGL1_003 DEM data, OpenStreetMap road data, and WorldPop/GP/100m/pop population density data; For the data required for runoff calculations, precipitation data comes from the ECMWF/ERA5 LAND/MONTHLY_AGGR dataset, NDVI data from COPERNICUS/S2_SR_HARMONIZED, and land cover data from ESA/WorldCover/v200; soil texture data from the projects/soilgrids-isric/clay_mean and projects/soilgrids-isric/sand_mean datasets in the SoilGrids database, and topographic factor data from the USGS/SRTMGL1_003 dataset.

Tree Species Data. Basic tree species data is primarily sourced from the CiTree Urban Tree Species Database[11], covering fundamental growth and morphological information such as scientific names, taxonomic status, family and genus information, crown spread, tree height, crown shape, growth rate, leaf characteristics, root system features, stress tolerance, and maintenance requirements; Functional trait data for tree species is primarily sourced from the <https://www.try-db.org> (TRY), including key indicators such as wood density, specific leaf area, leaf area, maximum height, growth habit, leaf dry matter content, leaf nitrogen content, and certain stomatal conductance values. For missing trait parameters in individual tree species, data were supplemented at the species, genus, or family level; where information remained unavailable, overall statistical values from the database were used to fill in the gaps. Additionally, to ensure consistency in species names and taxonomic information, scientific names were matched and standardized using plant taxonomic databases such as POWO and WFO.

2.3 Research Methods

A Method for Coupling Road Binary Classification with Land Use Scenarios.

Road classification utilizes eight core features. The functional gap layer includes NDVI, NPP, pollution pressure (PM_{2.5} + NO₂), runoff, and LST; the ecological constraint layer includes PET, soil pH, and salinity. Each variable is first standardized to the 0 – 1 range, and pollution pressure is calculated as an equally weighted combination of PM_{2.5} and NO₂. The primary functional category is determined based on the highest score among the five functional scores, classified as A (greening), B (carbon sequestration), C (pollution purification), D (stormwater management), or E (cooling); if the highest score is below the threshold T, it is classified as F

(comprehensive maintenance). Here, T is defined as the 60th percentile of the sample distribution, corresponding to the 60th percentile of the maximum functional scores for road samples (0.694263).

The ecological constraint index is calculated using a weighted combination: the weights for salinity, pH, and PET are 0.40, 0.35, and 0.25, respectively, and constraint levels are classified into low, medium, and high (Levels 1, 2, and 3) based on percentiles. Since there was a tie in the sample (P40 and P70 both had a value of 0.587318), percentile ranking was used to ensure that all three levels could be distinguished (40%/30%/30%). This resulted in 18 categories of “functional gap type–constraint level” codes (A-1 to F-3).

Based on this, the 2025 land use raster was overlaid. The dominant land use types along roads were extracted using raster mode and grouped into four categories: built (built-up areas), park (trees/grasslands/shrublands), agri (farmland and bare ground), and waterside (water bodies and floodplain vegetation). The proportion of each of the four scenarios was calculated for each final class, and a scenario-weighted suitability score was constructed (i.e., the suitability scores for each scenario were weighted by the proportion of that scenario within the road category). This approach avoids “selecting trees based on a single scenario” and ensures that the final recommendations reflect both the ecological shortcomings of the road and the actual land-use structure.

Method for screening suitable tree species. Subsequently, traits in the tree species database are mapped to numerical scores ranging from 0 to 1, including drought tolerance, heat tolerance, late frost tolerance, salt tolerance, waterlogging tolerance, maintenance requirements, invasion risk, and toxicity. Cold hardiness zones were assigned values in segments: zone ≤ 7.5 was assigned 1.0, $7.5 < \text{zone} \leq 8.0$ was assigned 0.6, and zone > 8.0 was assigned 0.2 (missing values were assigned 0.6); soil pH suitability was calculated based on the “overlap between the species’ pH range and the study area’s pH range.” The hard screening criteria for northern suitability are: $\text{hardiness_fit} \geq 0.6$, $\text{drought_tol} \geq 0.5$, $\text{frost_tol} \geq 0.5$, $\text{invasion_safe} \geq 0.5$, and $\text{tox_safe} \geq 0.4$. Based on these criteria, a comprehensive northern suitability score is calculated to rank the candidate tree species.

AHP + Entropy Weighting Method + TOPSIS Integrated Decision-Making. To quantitatively optimize tree species allocation schemes for different road categories, this study establishes an integrated decision-making framework combining AHP, the entropy weighting method, and TOPSIS. Six evaluation criteria are established: primary function, multi-function, constraint fit, land-use mix fit, management, and ecology. These represent, respectively, a tree species’ ability to address shortcomings in a road’s primary function, its performance in multifunctional synergy, its adaptability to PET/pH/salinity constraints, its suitability for mixed land-use scenarios, management safety, and ecological added value. AHP is used to reflect a priori preferences for different road types:

For non-F-class roads, the weight vector is as follows:

$$[9, 4, 2 + 2L, 5, 4, 2] \quad (1)$$

Class F roads use:

$$[5, 6, 4 + L, 5, 4, 4] \quad (2)$$

Here, $L \in \{1, 2, 3\}$ represents the constraint level. After vector normalization, the subjective weights w^{AHP} are obtained.

To reduce bias in subjective weighting, the entropy weighting method is further employed to calculate the objective weights. Let x_{ij} denote the normalized value of the i -th candidate tree species for the j -th criterion, then

$$p_{ij} = \frac{x_{ij}}{\sum_i x_{ij}}, \quad e_j = -k \sum_i p_{ij} \ln p_{ij}, \quad d_j = 1 - e_j, \quad w_j^{ENT} = \frac{d_j}{\sum_j d_j} \quad (3)$$

The combination weights were ultimately obtained using equal-weighting:

$$w_j = 0.5 w_j^{AHP} + 0.5 w_j^{ENT} \quad (4)$$

Based on this, the TOPSIS method is used to calculate the distances between the candidate tree species and the positive and negative ideal solutions, and the proximity:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (5)$$

As a comprehensive evaluation metric, a higher C_i value indicates greater overall suitability of a tree species. Based on this, this paper ranks candidate tree species for various road types and identifies the species with the highest proximity as the optimal species for the corresponding final class. It also outputs the top-N candidate results to support scheme comparison and sensitivity analysis.

3 Results and Analysis

3.1 Road Classification Results

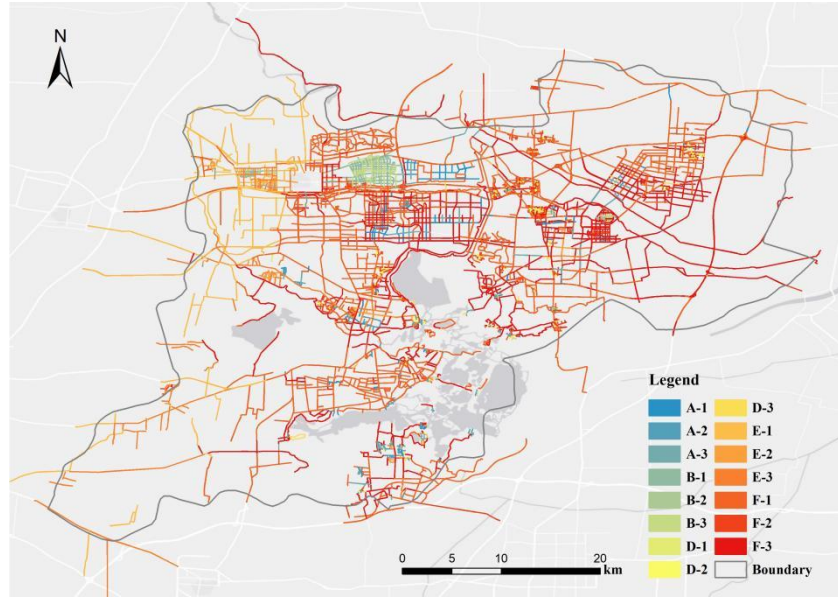


Fig 2. Map of Road Classification Results.

The results of road classification in the study area reveal a relatively distinct spatial pattern (see Fig. 2). Overall, Class F roads have the widest distribution and constitute the predominant type in the study area's road network. Among these, F-3 roads are primarily concentrated in the densely populated road networks of the central-eastern, eastern, and southern regions, forming a continuous distribution around *Baiyangdian* Lake; F-2 roads have the second-widest distribution and are mainly found on roads in the central, southeastern, and southern peripheries; F-1 roads are relatively scarce, mostly scattered across local sections in the central region and around the lake area. Class E roads also account for a significant proportion, primarily distributed in the western, southwestern, and northern and eastern peripheral areas of the study area. Among these, E-3 roads exhibit the most continuous distribution, E-2 roads are mainly found in the central-western and southern peripheries, while E-1 roads are mostly scattered along the western periphery and in local sections of the southwestern roads.

In contrast, Class A and Class B roads have a smaller distribution range and generally exhibit localized clustering. Class A roads are primarily distributed in the central-northern and northern regions, as well as on a few sections around *Baiyangdian* Lake. Among these, A-1 and A-2 are mainly concentrated in several contiguous road networks in the central-northern region, while A-3 is the least numerous, appearing only sporadically on individual roads; Category B roads are

primarily concentrated in the central-northern region, forming several block-like clusters, with B-1, B-2, and B-3 all being particularly prominent in the densely populated road areas of the north. Category D roads are generally fewer in number and are mainly scattered across the western, southwestern, and lake-adjacent regions, as well as on isolated sections in the central area. Among these, D-3 is relatively more common, while D-1 and D-2 appear only on a few scattered road segments. Overall, the northern and central-eastern parts of the study area exhibit a greater variety of road types and a denser spatial distribution, while the southwestern region and the area surrounding Baiyangdian Lake are characterized primarily by peripheral extension roads and localized clusters.

3.2 Tree Species Ranking and Optimization Capabilities

The TOPSIS ranking results indicate that the dominant tree species for different road categories show a high degree of consistency, though some variations still exist across different functional types and constraint levels. Overall, the top-ranked tree species for Class A, Class E, and Class F roads are quite similar, with *Diospyros lotus*, *Castanea sativa*, *Prunus avium*, *Ulmus glabra*, and *Alnus incana* ranking highly across multiple categories; Among Class B roads, *Ulmus glabra* ranks most consistently and remains near the top overall; Class D roads exhibit greater independence, with *Carya glabra* ranking first in D-1, D-2, and D-3.

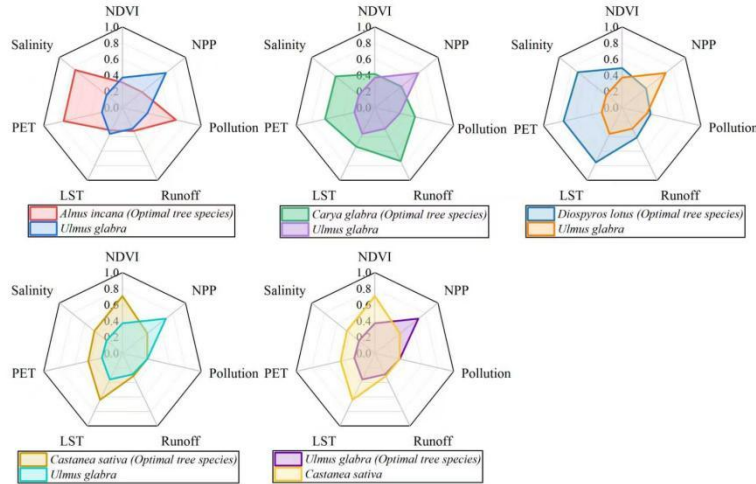


Fig 3. Radar chart showing the optimization capabilities of the optimal tree species for each road category

Specifically, the top four tree species in A-1 and A-2 were identical: *Castanea sativa*, *Diospyros lotus*, *Prunus avium*, and *Ulmus glabra*; the top three in A-3 were *Diospyros lotus*, *Prunus avium*, and *Castanea sativa*. The ranking results for B-1 and B-2 are also highly consistent, with *Ulmus glabra* ranking first in both, followed by *Castanea sativa* and *Amelanchier arborea*; in B-3, *Diospyros lotus* rose to first place,

with *Castanea sativa* and *Prunus avium* ranking second and third, respectively. The ranking results for the three levels in Class D were the most consistent, with the top three being *Carya glabra*, *Diospyros lotus*, and *Ulmus glabra*. The top four rankings in E-1 and E-2 were consistent, namely *Diospyros lotus*, *Castanea sativa*, *Ulmus glabra*, and *Prunus avium*; In E-3, *Diospyros lotus* remained in first place, with *Prunus avium* and *Castanea sativa* ranking second and third, respectively. The top three rankings were consistent between F-1 and F-2, in the order of *Diospyros lotus*, *Ulmus glabra*, and *Castanea sativa*; in F-3, *Diospyros lotus* retained first place, followed closely by *Prunus avium*, *Alnus incana*, and *Castanea sativa*.

The radar chart further shows (see Fig. 3) that different dominant tree species exhibit significant differences in their performance across the various optimization dimensions. *Alnus incana* scored higher in the pollution removal, PET, and salinity tolerance dimensions, with particularly strong performance in PET and salinity tolerance; *Carya glabra* performed well in runoff, PET, and salinity tolerance, with a relatively balanced overall distribution; *Castanea sativa* stood out in the NDVI and LST cooling dimensions; *Diospyros lotus* performs most notably in LST cooling, PET, and salinity tolerance, while also maintaining a high level in the NDVI dimension; *Ulmus glabra*'s dominance is primarily concentrated in the NPP dimension, and this indicator is significantly higher than in the other dimensions. Overall, the tree species ranking at the top of each category exhibit relatively distinct differences in functional performance, and their capability distributions generally align with the ranking results of their respective categories.

3.3 Optimization Results Based on High-Ranking Tree Species

In terms of temperature mitigation, *Diospyros lotus* has the highest optimization potential, with a LST calculation capacity index of 1.521186. In the tree species parameter table, this species is classified as a medium-sized tree with an average crown radius of approximately 10 m, an average height of approximately 10 m, and a maximum height of approximately 12 m. It typically has a rounded crown shape and exhibits high drought tolerance, with a soil pH tolerance range of approximately 5.5–8.0. Under the scenario of “Optimal tree species combination for planting in a 15-meter roadside buffer zone,” the mean temperature in the study area decreased from 37.162403 to 35.685158, a net decrease of 1.477245, representing a reduction of approximately 3.98%.

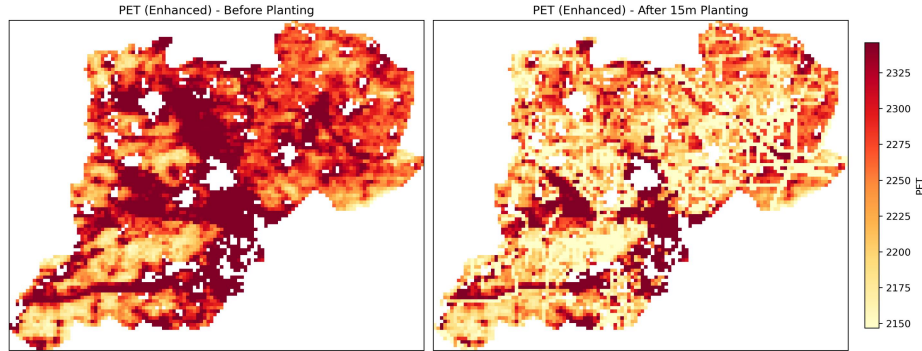


Fig 4. Before-and-after PET comparison of planting the optimal tree species in the 15-meter buffer zone

In terms of Potential Evapotranspiration mitigation (see Fig. 4), *Alnus incana* performed best, with a PET calculation index of 7.040000%. This species has an average crown radius of approximately 8 m, an average height of approximately 15 m, and a maximum height of approximately 20 m. Its crown shape is a composite of broad, oval, conical, and rounded types; its cold hardiness zone reaches Zone 2, its de-icing salt tolerance rating is “good,” its soil pH tolerance range is approximately 5–8, and its salt tolerance index (salt tol) is 1.0. Simulation results show that the mean PET decreased from 2313.629150 to 2182.078857, a net decrease of 131.550293, representing a reduction of approximately 5.69%.

In terms of NPP enhancement, *Ulmus glabra* showed the most significant improvement, with an NPP enhancement index of 14.783429%. This species is a typical tall tree, with an average crown radius of approximately 15 m, an average height of approximately 35 m, and a maximum height of approximately 40 m, indicating a large canopy size; Its soil pH tolerance range is approximately 5.5–7.5, but its salt tolerance is relatively low, with a salt_tol value of 0.2. Under the corresponding scenario, the mean NPP increased from 295.812707 to 321.776276, representing a net increase of 25.963569, or approximately 8.78%.

Regarding pollution index mitigation (see Fig. 5), the optimal tree species is also *Alnus incana*, with a pollution mitigation capacity index of 14.297441%. This species has a high particulate matter adsorption capacity and good tolerance to de-icing salts, with a salt_tol value of 1.0; its average crown radius is approximately 8 m, and its average tree height is approximately 15 m. Scenario simulations indicate that the mean pollution index decreased from 0.499485 to 0.457136, a net decrease of 0.042350, representing a reduction of approximately 8.48%.

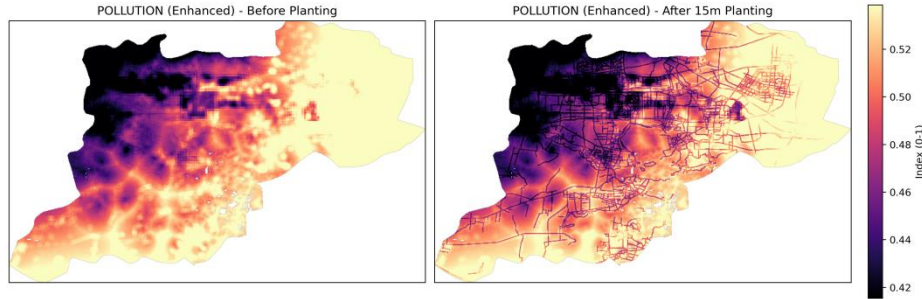


Fig 5.Before-and-after Pollution comparison of planting the optimal tree species in the 15-meter buffer zone

4 Discussion

The results of this study indicate that the ecological optimization needs of roads in the Xiong'an New Area exhibit distinct spatial variations, making it inappropriate to use a single tree species for uniform planting across the entire area. In the road classification results, Class F and Class E roads constitute the majority, while Class A, B, and D roads are primarily distributed in localized clusters. This indicates that the road network in the study area generally prioritizes comprehensive maintenance and thermal environment improvement, with only certain sections exhibiting prominent deficiencies in specific functions. Consequently, street tree planting should focus on zone-specific and category-based optimization rather than uniform replacement based on a single objective.

The tree species ranking results further illustrate that while there is some overlap in the dominant tree species across different road categories, their dominant characteristics vary. *Diospyros lotus*, *Castanea sativa*, *Ulmus glabra*, *Prunus avium*, and *Alnus incana* consistently rank among the top in multiple categories, indicating that these species possess good overall adaptability to the environmental conditions of the study area; Meanwhile, the consistent top ranking of *Carya glabra* in Category D roads indicates that tree species selection exhibits stronger specialization under specific functional objectives. In other words, this study did not identify a single “optimal tree species,” but rather a combination of dominant tree species with different functional emphases.

The results from the radar plots and scenario simulations further corroborate this finding. *Diospyros lotus* demonstrates strong performance in terms of LST, PET, and salt tolerance, making it suitable as a versatile tree species that balances thermal environment improvement with stress tolerance; *Ulmus glabra*'s strengths are primarily concentrated in the NPP dimension, making it more suitable for carbon sequestration and biomass enhancement functions; *Alnus incana* excels in pollution reduction and PET mitigation, demonstrating strong environmental regulation capabilities; *Carya glabra*, meanwhile, exhibits a more balanced performance across runoff, PET, and salinity tolerance dimensions. This indicates a clear functional

division among different tree species, suggesting that road tree species optimization is best approached through a “backbone species + functional supplementary species” configuration strategy.

Methodologically, this study incorporates road functional gaps, ecological constraint levels, and land-use scenarios into the tree species selection process, ensuring that the recommended results not only address ecological deficiencies along roads but also account for environmental adaptability and spatial context. Consequently, the results are closer to actual planting requirements than rankings based on a single functional criterion. At the same time, it should be noted that this study remains subject to factors such as differences in the spatial scales of multi-source data, the need to fill in missing trait parameters for some tree species, and simplifications in scenario modeling. Further validation through field surveys and long-term monitoring is therefore required.

5 Conclusion

Taking the Xiong'an New Area as the study area, this paper constructs an ecological optimization framework for roadside trees in road spaces. By integrating the identification of road functional gaps, the classification of ecological constraints, the coupling of land-use scenarios, and the screening of suitable tree species with a comprehensive decision-making approach combining AHP, the entropy weighting method, and TOPSIS, the study achieves the quantitative identification and differentiated allocation of optimal tree species across different road categories. The results indicate that road ecological demands in the Xiong'an New Area exhibit significant spatial heterogeneity, with Class F and Class E roads dominating the landscape, while Class A, B, and D roads are primarily distributed in localized clusters; Although high-ranking tree species across different road categories exhibit some overlap, they demonstrate distinct functional differences. Specifically, *Diospyros lotus* performs well in temperature mitigation and stress tolerance; *Ulmus glabra* excels in enhancing NPP; *Alnus incana* shows strong potential for PET and pollution reduction; and *Carya glabra* is better suited for road types with high stormwater management requirements. Overall, the results indicate that there is no single optimal tree species suitable for all road types; instead, tree selection should be based on the ecological weaknesses of the road, environmental constraints, and spatial context to implement a categorized configuration.

The significance of this study lies in transforming the traditionally experience-driven problem of roadside greening configuration into a quantitative decision-making problem focused on multi-objective ecological benefits and multiple environmental constraints. It provides an actionable technical pathway for optimizing roadside greening in the Xiong'an New Area and offers a methodological reference for other new urban areas seeking to implement refined street tree configurations. At the same time, due to differences in the spatial resolution of multi-source remote sensing and reanalysis data, the interpolation of certain tree species trait parameters, and the simplification of scenario simulation conditions, the results of this study still

carry a degree of uncertainty. In the future, tree species allocation schemes can be continuously refined and optimized by integrating field surveys, long-term monitoring data, and practical constraints such as road cross-sections, underground utilities, and maintenance costs, thereby enhancing the practical value of these findings in urban greening planning and ecological management.

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